Program : **Diploma in Polymer Technology**	
Course Code : **5096**	Course Title: **Quality Assurance And Testing Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To get an exposure for the students in raw rubber testing, in process tests and rubber vulcanizate testing.
- To prepare the student to be confident enough to handle any problems in dry rubber industry which will definitely assure quality of that product.
- To get a clear idea on the importance of quality assurance and quality control in rubber industries.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of Analytical Chemistry		Applied Chemistry	1
		Applied Science Lab-Chemistry	1
		Polymer Science Lab	3
Specification of ISNR grades.		Natural rubber production	3
Rubber Compounding		Technology of elastomers	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Perform various specification tests of Natural rubber as per IS 4588	29	Applying
CO2	Carry out the in-process tests of rubber compound.	9	Applying
CO3	Perform tests on rubber vulcanizates.	5	Applying
	Lab Exam	2	

CO -PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3		3			
CO3	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Perform various specification tests of Natural rubber as per IS 3708**		
M1.01	Estimate dirt content of raw natural Rubber	6	Applying
M1.02	Determine the ash content of raw natural rubber	6	Applying
M1.03	Determine volatile matter of raw natural Rubber.	3	Applying
M1.04	Estimate nitrogen and total protein content of raw natural Rubber.	6	Applying
M1.05	Determine acetone extract of raw natural rubber.	6	Applying
M1.06	Determination of Po & PRI of raw natural rubber	2	Applying
	Lab Exam I	1	
CO2	**Carry out the in-process tests of rubber compound.**		
M2.01	Estimate carbon black content of rubber compound	6	Applying
M2.02	Determination of density of rubber compound	3	Applying
CO3	**Perform tests on rubber vulcanizates**		
M3.01	Determine swelling behaviour of rubber vulcanizates.	5	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
1	Rubbery Materials and their Compounds - J.A. Brydson (Publisher - Springer Netherlands , 1998)
2	Rubber Technology – Maurice Morton (Publisher – Springer, 1973)
3	Training Manual -RRII, Kottayam
4	Physical testing of Rubbers – R.P. Brown
5	Rubber Standards - ASTM International
6	Rubber and rubber products - ISO/TC 45

Online Resources:

Sl.No	Website Link
1	https://archive.org
2	http://www.indiannaturalrubber.com
3	https://www.astm.org
4	https://pubs.acs.org
5	https://www.iso.org